

PulmoScan: AI for Lung Nodule Detection

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Date: April 29, 2025

Revolutionizing Lung Cancer Detection with AI

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Project Overview

PulmoScan is an innovative project that uses artificial intelligence (AI) to aid in the detection and classification of lung nodules (small growths in the lungs). By assisting healthcare professionals in identifying lung problems, including cancer, our project aims to enable early detection, which can save lives through faster access to treatment.

PulmoScan analyzes medical images—such as chest X-rays and CT scans—using advanced AI models trained to:

- **Detect Lung Nodules:** Precisely locate nodules in the lungs.
- **Classify Images:** Categorize nodules as normal, benign (non-cancerous), or malignant (cancerous).
- **Identify Lung Cancer Types:** Recognize specific cancer types, including:
 - Squamous cell carcinoma
 - Adenocarcinoma
 - Large cell carcinoma

Business & Data Science Goals

Business & Data Science Goals	
Business Goals	Data Science Goals
Detect lung nodules early to improve survival rates and treatment success.	Train AI models to identify nodules with high accuracy.
Make diagnoses more accurate by reducing human errors.	Use advanced AI techniques to reduce false positives and false negatives.
Reduce doctors' workload so they can focus on complex cases.	Develop automated nodule detection and classification.
Make AI tools easy to use in hospitals and clinics.	Create a user-friendly system with visual outputs and reports.
Support research and innovation in medical imaging.	Build AI models that can be tested, improved, and used by other researchers.
Save costs for hospitals by reducing unnecessary tests and follow-ups.	Improve efficiency of AI models to work with fewer resources.
Help doctors make better treatment decisions by identifying cancer types quickly.	Train models to classify nodules into normal, benign, or cancerous categories.

Table 1 : Business & Data Science Goals table

Business Value

3.1 For Hospitals and Clinics

- Faster and more accurate diagnoses.
- Improved patient outcomes through earlier treatment.

3.2 For Medical Professionals

- Reduced workload and error rates with AI support.
- Better resource allocation in radiology departments.

3.3 For Healthcare Systems

- **Cost Reduction:** Save on expensive follow-up diagnostics through higher initial accuracy.

Key Performance Indicators (KPIs)

4.1 Detection Accuracy

- **Precision & Recall:** Measures how well the AI detects real nodules while avoiding mistakes.
- **Intersection over Union (IoU):** Checks if the AI correctly marks the exact location of a nodule.

4.2 Classification Performance

- **Accuracy:** Measures how often the AI correctly labels nodules as normal, benign, or cancerous.
- **F1-Score:** Balances precision and recall to ensure reliable classification of different cancer types.

4.3 Efficiency and Time-Saving

- **Processing Time:** Measures how quickly AI analyzes an image.
- **Reduction in Doctor Review Time:** Tracks how much time doctors save per case.

4.4 Real-World Impact

- **Early Cancer Detection Rate:** Measures how many cancers are detected at an early stage compared to before using AI.
- **Adoption Rate:** Checks how many hospitals and doctors start using PulmoScan.
- **Cost Savings:** Tracks how much hospitals save by reducing unnecessary tests and follow-ups.

Relevant Sustainable Development Goals (SDGs)

- **SDG 3: Good Health and Well-being** – Enables early and accurate detection of lung cancer, improving survival rates and reducing workload on radiologists.
- **SDG 9: Industry, Innovation, and Infrastructure** – Advances medical imaging through AI technology, optimizing processes for efficiency.
- **SDG 12: Responsible Consumption and Production** – Uses optimized AI models to reduce computing costs and energy consumption.
- **SDG 17: Partnerships for the Goals** – Encourages collaboration between healthcare providers, researchers, and industry.

Risks (Challenges to Overcome)

- **Data Quality:** If the datasets are incomplete or unbalanced, the AI model may not work well for all patients.
- **Regulatory Challenges:** AI tools in healthcare must pass strict approval processes before they can be used in hospitals.
- **Adoption Barriers:** Some doctors and medical staff may be hesitant to trust or use AI-based diagnosis tools.

Assumptions (What We Expect to Work)

- **Reliable Data:** The datasets used (LUNA16, TCIA, NIH) accurately represent real-world cases.
- **Good Model Performance:** AI models will work well on new, unseen medical images, not just the ones they were trained on.
- **Easy Integration:** PulmoScan can be smoothly added to hospital systems without major technical issues.

Stakeholders

8.1 Primary Stakeholders

- **Healthcare Providers:** Radiologists, oncologists, and diagnostic teams directly using PulmoScan.
- **Patients:** Beneficiaries of earlier and more accurate diagnoses.

8.2 Secondary Stakeholders

- **Hospital Administrators:** Seek cost-efficient diagnostic solutions.
- **AI Researchers:** Interested in scientific contributions and model innovations.
- **Industry Partners:** Collaborate on data provision or product commercialization.

Data Acquisition

A diverse dataset is essential for training robust AI models. The following datasets were used:

- **LUNA16 Dataset:** Specially designed for the detection of pulmonary nodules. This dataset ensures that AI models can effectively recognize and classify lung nodules, improving diagnostic accuracy.
- **TCIA (The Cancer Imaging Archive):** Contains medical images for various types of cancers, providing a comprehensive dataset for training AI models in diverse real-world scenarios.
- **NIH Chest X-ray Dataset:** Provides annotated chest X-rays with associated diagnostics, helping AI models learn from labeled medical images to improve detection accuracy.
- **IQ-OTHNCCD Lung Cancer Dataset:** Collected from the Iraq-Oncology Teaching National Center for Cancer Diseases (IQ-OTH/NCCD), this dataset includes CT scans of patients diagnosed with lung cancer at different stages, as well as healthy subjects, ensuring a well-balanced dataset for training.

Each dataset serves a specific purpose in improving AI performance, ensuring accuracy in both detection and classification.

Data Preparation

To enhance model robustness, various preprocessing techniques were applied to different datasets:

10.1 LUNA16 Dataset

Injecting Gaussian and Poisson noise helps our model generalize better, improving its accuracy in real-world medical imaging, we incorporate:

- **Gaussian Noise:** Simulates sensor noise from CT scans, improving resilience to scanning artifacts.
- **Poisson Noise:** Mimics low-dose CT scans, ensuring AI models perform well even with reduced radiation images.

10.2 Chest CT-Scan Images Dataset

Normalization and resizing were applied to make model training more stable, faster, and consistent by standardizing pixel values and ensuring all CT-Scan images fit the model architecture for efficient, uniform processing.

- **Resize:** Adjusts all CT-Scan images to a uniform size, ensuring compatibility with the model architecture.
- **Normalization:** Scales pixel values to a standard range, making model training more stable, faster, and consistent.

10.3 TCIA (Cancer Imaging Archive)

Combining Motion Blur Speckle Noise ensures AI models can accurately detect lung nodules in real-world medical scans.

- **Motion Blur:** Simulates blurring caused by patient movement during scanning.
- **Speckle Noise:** Represents distortions from ultrasound or low-quality imaging in some cancer scans.

10.4 Applied to All Datasets

- **Normalization:** Scales the data to a standard range, typically between 0 and 1, ensuring consistency across images and improving model performance.
- **Rotation:** Rotates the image by a certain angle to augment the dataset and help the model recognize features from different orientations.
- **Flipping:** Flips the image horizontally or vertically to create mirror versions, enhancing model robustness to different perspectives.
- **Cropping:** Cuts out a region of interest from the image, focusing on important areas and improving model efficiency and accuracy.

These techniques help AI models adapt to real-world imaging challenges, improving detection accuracy.

Modeling Phase

In this phase, we developed and trained deep learning models to classify lung nodules from Chest CT-Scan images. The modeling is divided into two parts: primary classification and secondary classification, each using specific architectures based on task complexity and resource considerations.

11.1 Primary Classification

The primary classification task aims to distinguish between CT-Scan images containing lung nodules and those without. Two convolutional neural network (CNN) architectures were selected:

- **MobileNetV2:** Used because it offers efficient, low-computation feature extraction. It leverages transfer learning from ImageNet for resized lung nodule images (224×224), and generalizes well on small medical datasets, reducing the risk of overfitting.

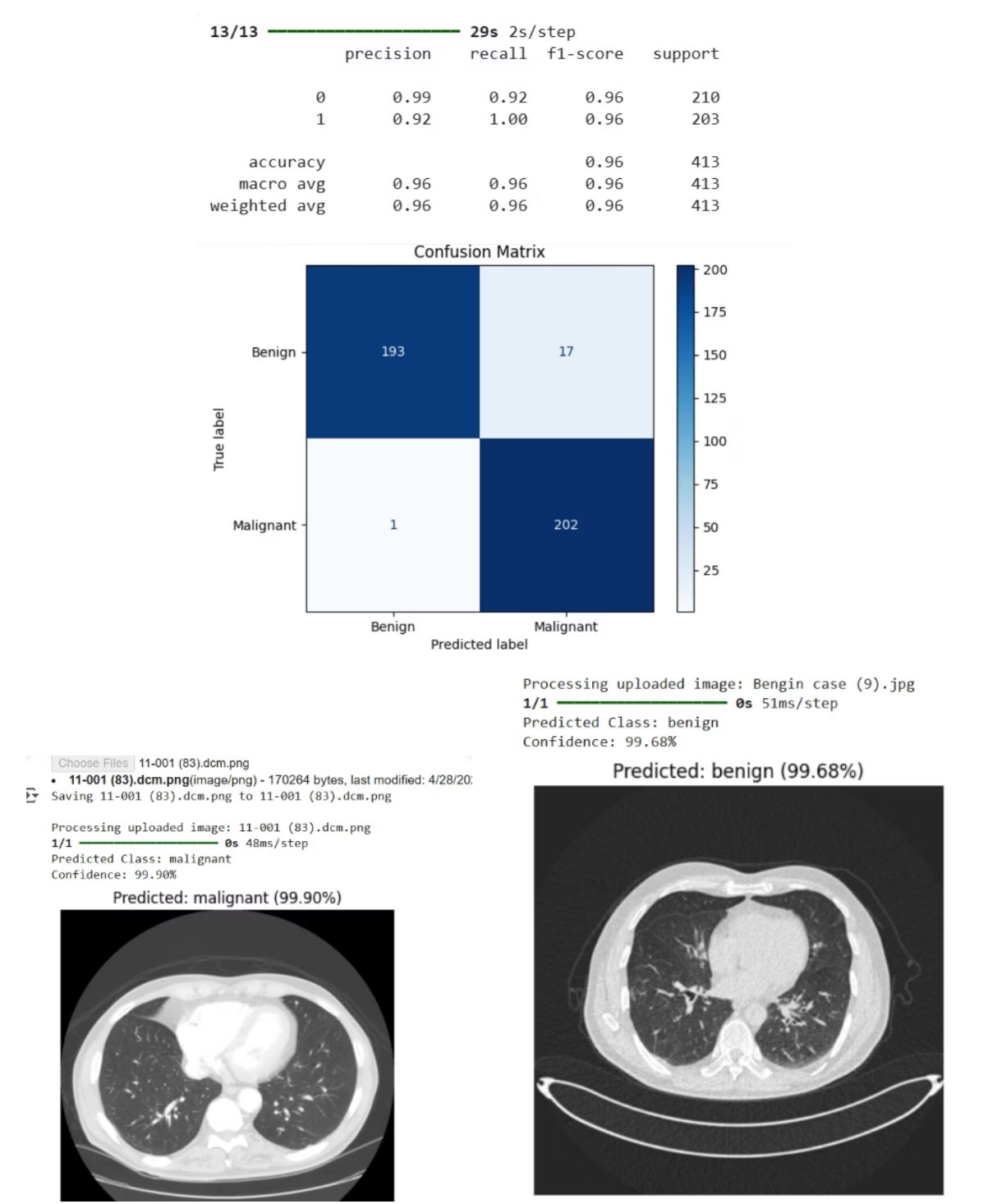


Figure 1: MobileNetV2 model visualizations and outputs.

- **DenseNet121:** Chosen due to its dense connectivity, which enables rich feature extraction and robust learning across variable lung nodule appearances. Its effective use of transfer learning from ImageNet helps achieve high classification accuracy.

13/13 ————— 13s 1s/step - accuracy: 0.9633 - loss: 0.0810
 Test Loss: 0.11551424860954285
 Test Accuracy: 0.9491525292396545
 13/13 ————— 23s 993ms/step

	precision	recall	f1-score	support
0	0.97	0.93	0.95	210
1	0.93	0.97	0.95	203
accuracy			0.95	413
macro avg	0.95	0.95	0.95	413
weighted avg	0.95	0.95	0.95	413

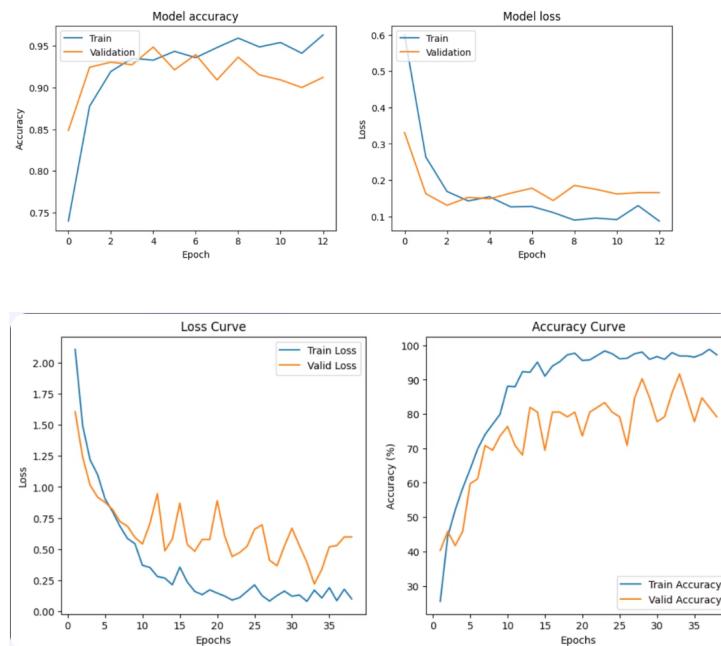
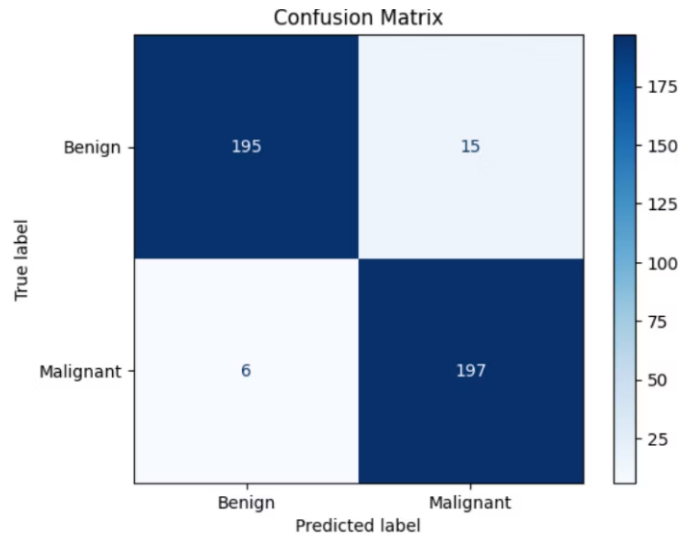


Figure 2: DenseNet121 model visualizations and outputs.

11.2 Secondary Classification

The secondary classification task involves more fine-grained categorization of nodules. Four different deep learning models were employed to enhance learning capacity and accuracy:

- **ResNet18:** Selected for its residual connections that allow very deep networks to learn effectively

without suffering from vanishing gradients, making it ideal for capturing complex lung nodule features.

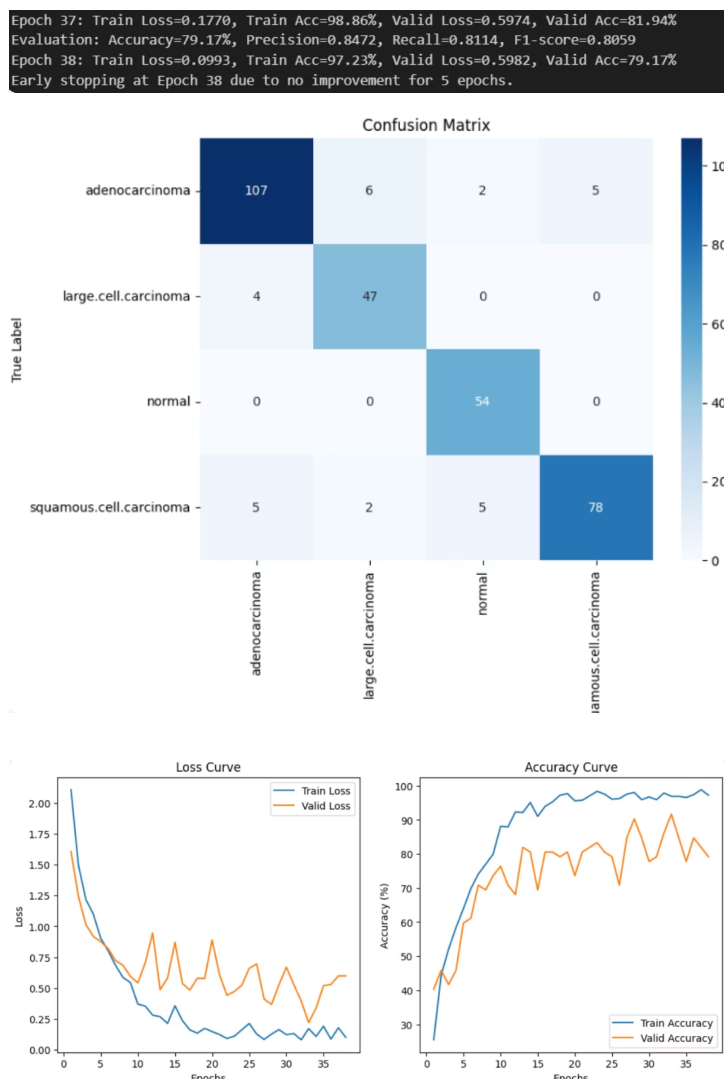


Figure 3: ResNet18 model visualizations and outputs.

- **DenseNet121:** Applied again due to its feature reuse and robust learning properties, helping the model capture subtle differences in lung nodule appearances with fewer computations.



Figure 4: DenseNet121 model visualizations and outputs.

- **EfficientNetB0**: Chosen for its ability to achieve high classification accuracy while minimizing computational cost, making it highly efficient for classifying lung nodules in resource-constrained environments.

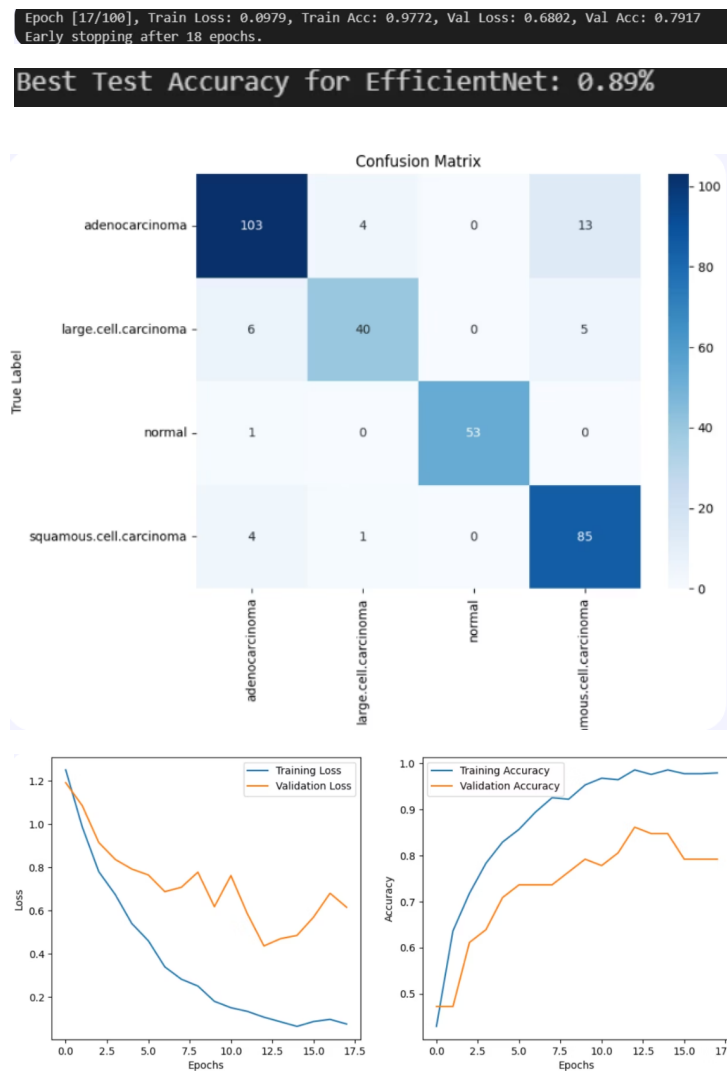


Figure 5: EfficientNetB0 model visualizations and outputs.

- **Xception:** Selected because its use of depthwise separable convolutions enables faster and more precise extraction of important features from lung CT-Scan images.

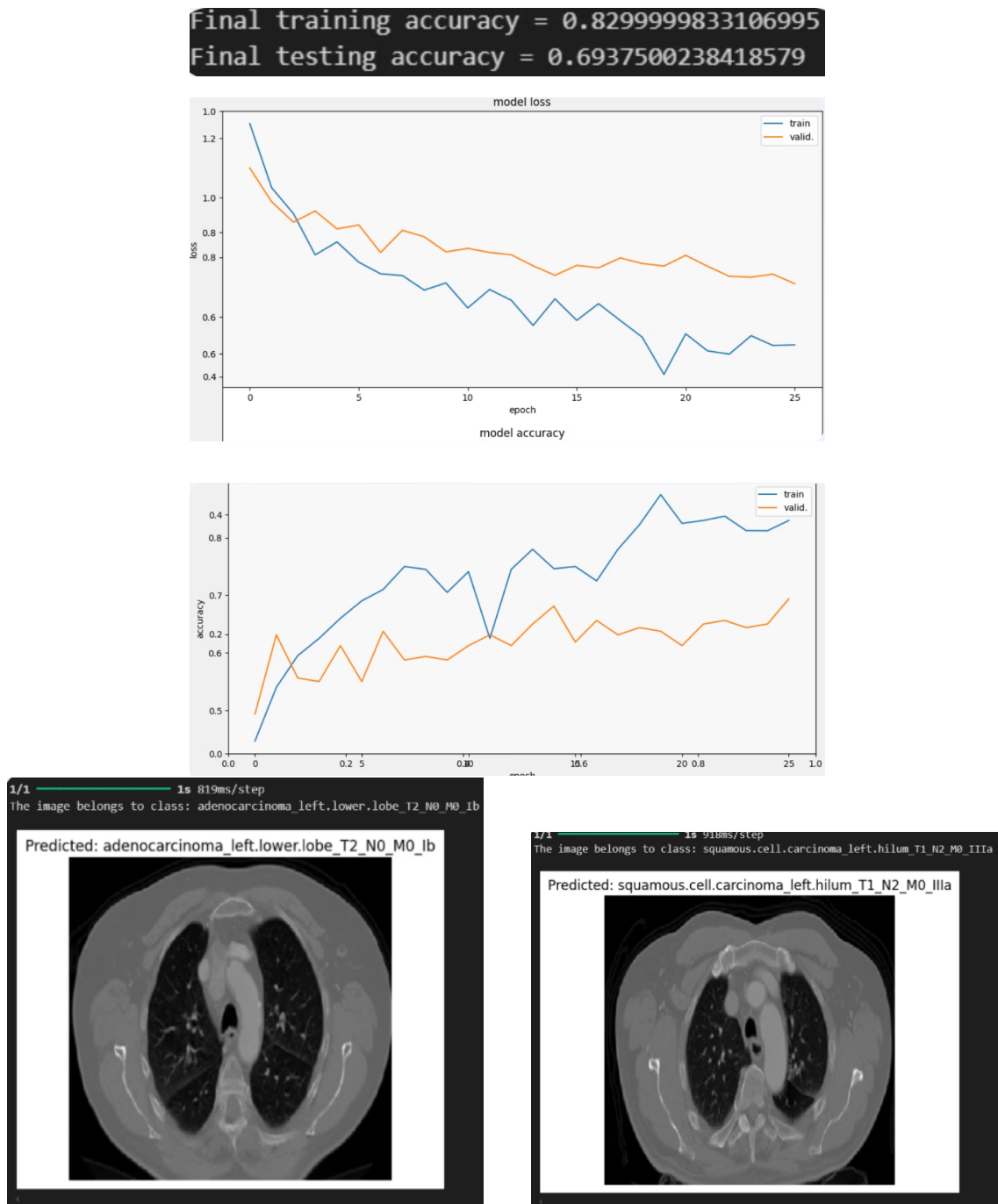


Figure 6: Xception model visualizations and outputs.

11.3 Pulmonary Nodule Detection

In addition to classification, we implemented a detection pipeline to localize lung nodules in Chest CT-Scan images using the **YOLOv8** (You Only Look Once version 8) architecture.

- **YOLOv8:** Selected for its speed and accuracy in real-time object detection, YOLOv8 is capable of detecting nodules of various sizes and shapes, making it ideal for pulmonary imaging applications.

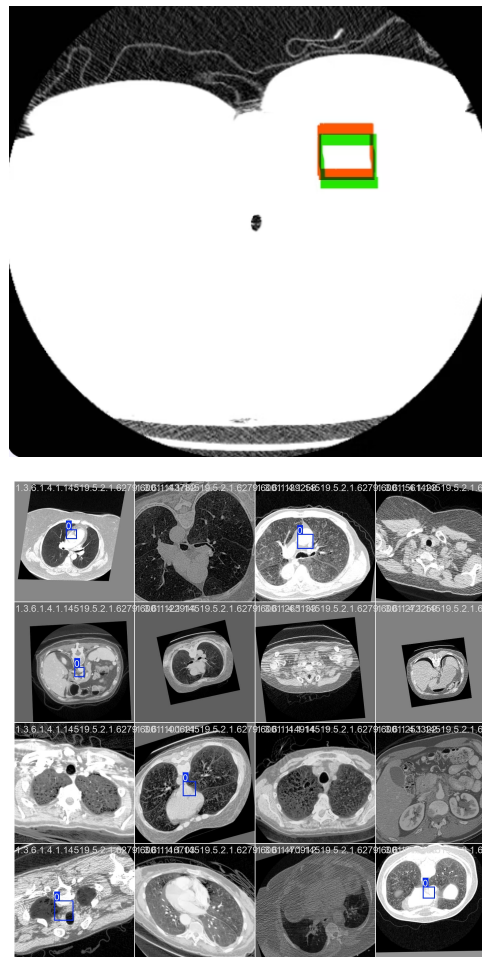


Figure 7: YOLOv8 detection outputs highlighting nodule localization on CT-Scan images.

Future Steps

AI Model Training:

- Deployemt and Scientific report .

Conclusion

PulmoScan revolutionizes lung disease diagnosis with AI-driven precision, enhancing accuracy, efficiency, and patient outcomes. By reducing workload and costs, it empowers radiologists with seamless integration and real-world impact. This innovative solution redefines medical diagnostics.